Likelihood-Based Model Selection in High-Dimensional Hidden Markov Models

Ghaleb Khalil
University of Washington
ghaleb@uw.edu

Statistical Methods in Computer Science Project Report

Project Scope - Summary

This study aims to investigate model order selection in high-dimensional Gaussian emission Hidden Markov Models (HMM) in high-frequency limit order book data and assess whether log-likelihood, AIC, and BIC provide reliable model order selection criteria in non-regular latent models.

The main question this study tries to answer is whether well-known limitations in Hidden Markov Models, such as non-identifiability and likelihood singularities and multimodality, lead to model over-selection and instability and fragmentation of underlying regimes when models are estimated by the EM algorithm. In this study, three hypotheses will be tested:

Hypothesis 1: Information criteria under-penalize model complexity. Hypothesis 2: Models selected by information criteria are unstable across different initial values and subsamples. Hypothesis 3: With a larger K, a small number of regimes are fragmented into statistically distinct yet economically redundant states.

Methodology - Summary

The entire hidden Markov model (HMM) pipeline was implemented de novo in Python using the NumPy and SciPy libraries. The code was written in accordance with existing references. The code performs the entire HMM pipeline, including forward-backward inference, Baum-Welch estimation, and model selection. The code was used for model fitting for $K = 1, \ldots, 20$, with multiple random initializations.

The experiments use high-frequency data from the Nasdaq TotalView ITCH feed. Two feature sets are used: the 3D and 20D representations. The data was standardized and clipped at 50,000 data points. For the HMMs, the log likelihood, AIC, and BIC are used for model selection. Other diagnostics include state alignment using the Hungarian algorithm, comparison of the transition probability matrices, comparison of the state sequences for disagreement, and the KL divergence of the emission distributions. The code was run on standard CPU hardware with the assumption of Gaussian emission distributions with diagonal covariance.

Results - Summary

The main conclusion is that the three criteria of log-likelihood, AIC, and BIC do not provide reliable guidance for choosing the number of hidden states in high-dimensional hidden Markov models. Although the criteria framework consistently worked, it was observed that all criteria favored more complex models. This is because the log-likelihood increases with K, while AIC and BIC monotonically decrease with K, favoring the largest model available, with no clear optimum, hence over-selection. High K models also tend to have fragmented and imbalanced state structures, with dominant but less probable states, as well as a large number of transitions. This also hinders interpretability. The drawback of the current study is that it only deals with Gaussian emission HMMs with diagonal covariance and a single data representation.

What was Easy - Summary

The more straightforward aspects were interpreting results and generating diagnostic plots. Producing likelihood curves, state distributions, and visualizations was relatively intuitive once models were fitted.

CSE 599 , Project Report.

What was Difficult - Summary

The main challenges were implementing the full HMM pipeline from scratch and handling its computational and numerical complexity. Ensuring stable forward–backward and EM procedures required careful use of multiple references. In addition, fitting models across multiple K values and interpreting high-K models with fragmented states was challenging.

1 Introduction

Hidden Markov Models (HMMs) are a class of probabilistic models widely used to capture regime-switching behavior in time series data. Since the seminal tutorial by Rabiner (1) and the comprehensive treatment by Cappé, Moulines, and Rydén (2), HMMs have become standard tools in applications such as speech recognition, signal processing, and finance. In financial settings, they are particularly useful for modeling high-frequency market dynamics, where latent states can represent changes in liquidity, order flow imbalance, or volatility regimes (5).

A central modeling decision in HMMs is the choice of the number of hidden states. Unlike parameters such as transition probabilities or emission distributions, the number of states determines the overall structure of the model. Choosing too few states leads to underfitting, while choosing too many can result in unstable or redundant regimes that are difficult to interpret. In practice, this model selection problem is typically addressed using likelihood-based criteria such as the Akaike Information Criterion (AIC) (3) and the Bayesian Information Criterion (BIC) (4).

However, applying these criteria to HMMs is theoretically problematic. As latent variable models, HMMs often violate the regularity conditions required by classical likelihood theory. In particular, their likelihood functions can exhibit non-identifiability, multimodality, and singularities, as is well known in mixture models (2; 8; 9). These issues undermine the theoretical guarantees of AIC and BIC, and their practical performance in such settings is not well understood. The problem becomes more severe in high-dimensional settings: as the dimension of the observations increases, Gaussian-emission HMMs may split a single underlying regime into multiple statistically distinct but structurally similar states, leading to over-partitioned and difficult-to-interpret models.

Despite their widespread use, there is limited empirical understanding of how AIC and BIC behave in high-dimensional financial applications such as limit order book data. In particular, it remains unclear whether these criteria can reliably identify a meaningful number of hidden states when the likelihood surface is highly non-regular and the latent structure is difficult to recover.

In this project, we provide an empirical investigation of likelihood-based model selection in Gaussian-emission HMMs applied to high-frequency market microstructure data. We show that classical information criteria systematically favor increasingly complex models, without identifying a clear or meaningful optimal number of hidden states. To better understand this behavior, we complement model selection curves with structural diagnostics, including state occupancy distributions, transition dynamics, and divergence measures between emission distributions.

Our findings suggest that, in high-dimensional non-regular settings, likelihood-based criteria behave more like complexity-seeking objectives than reliable model selection tools. These results highlight the need for caution when applying classical information criteria to modern high-dimensional HMMs and motivate the use of additional diagnostics for model assessment.

2 Scope of the Project

The goal of this project is to empirically evaluate whether likelihood-based model selection criteria, specifically log-likelihood, AIC, and BIC, provide reliable guidance for selecting the number of hidden states in high-dimensional Gaussian-emission Hidden Markov Models applied to limit order book data. The research question of this project was to investigate whether these information-based model selection rules are effective tools for hidden state dimensionality selection of non-regular, high-dimensional models of regime structures, or whether these rules are systematically prone to over-complex and unstable regime structures. The objective of this project was not to develop new asymptotic theory for these models or to propose new information-based model selection rules. Instead, the objective of this project was to implement, from the ground up, the complete model fitting pipeline and to investigate whether the well-known limitations of these models, such as non-identifiability, likelihood singularities, multimodality, and non-standard asymptotic theory, are reflected in these models' empirical performance on actual high-frequency financial data. The objective of this project was to focus narrowly on the following topics: Gaussian emission Hidden Markov Models, Maximum Likelihood estimation using the Expectation-Maximization algorithm, model selection over varying hidden state dimensions K, diagonal emission covariance structures, and a single asset class over controlled event time windows.

The objective of this project was not to investigate Bayesian nonparametric models, penalized likelihood methods, or alternative inference techniques.

2.1 Addressed Claims/Hypotheses from the Original Paper

In order to make the assessment empirically testable, the following hypotheses are proposed.

1. **Hypothesis 1: Likelihood Over-Selection**
   In high-dimensional hidden Markov models estimated on limit order book data, log-likelihood, AIC, and BIC might not sufficiently penalize model complexity. This could lead to further improvements even for increasing numbers of states K, which do not represent meaningful additional structure. We define over-selection as follows: AIC or BIC continue to decrease beyond the point where newly added states show: considerable overlap in their emission distributions (measured by pairwise KL divergence), very similar transition behavior, and strongly underoccupied effective state distributions.
2. **Hypothesis 2: Instability Across Fits**
   AIC- or BIC-based model selection might result in unstable models. The optimal number of states or the selected structure might depend on: different random initializations of the EM algorithm and different temporal windows of the data. Instability is checked by comparing: (i) state sequences, (ii) transition matrices, and (iii) evidence for splitting or merging of regimes across fits. If models that are known to be equivalent produce significantly different results, this would indicate that information-criterion-based model selection does not correspond to a stable structure.
3. **Hypothesis 3: Fragmentation of Hidden Regimes**
   With increasing numbers of hidden states, the estimated HMM might divide a small number of underlying market regimes into a larger number of hidden states. This would be reflected in: strongly irregular state occupancy distributions, several states with very low effective probability, and transitions between multiple states on short time scales. If these phenomena are observed for high-dimensional models, they would indicate that adding more states does not reveal new economic regimes but instead fragments representations of the same underlying structure.

3 Methodology

The present research relies on a theory-informed empirical approach to test the behavior of classical likelihood-based model selection criteria in high-dimensional Gaussian emission Hidden Markov Models (HMMs). The goal is not to improve any specific inference method but to test whether well-known non-regular patterns in latent variable models appear empirically as instability or over-selection when used in realistic market microstructure models. All HMM-related procedures are implemented in Python independently according to the descriptions in (1; 2), along with specific procedures designed to test model comparison and stability.

3.1 Model and Estimation

Gaussian-emission HMM is considered with K hidden states. Let $\{Z_t\}_{t=1}^{T}$ denote the latent Markov chain, where $Z_t \in \{1, \dots, K\}$ and $\mathbb{P}(Z_t = j \mid Z_{t-1} = i) = A_{ij}$, with initial distribution $\pi_i = \mathbb{P}(Z_1 = i)$. The transition matrix $A \in \mathbb{R}^{K \times K}$ satisfies row-stochastic constraints. Conditional on the latent state,

$$X_t \mid Z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k),$$

where $X_t \in \mathbb{R}^d$, $\mu_k \in \mathbb{R}^d$, and $\Sigma_k \in \mathbb{R}^{d \times d}$. The parameter set is

$$\theta = \big(\pi, A, \{\mu_k, \Sigma_k\}_{k=1}^{K}\big).$$

The observed-data log-likelihood is

$$\ell(\theta) = \log p_\theta(x_{1:T}) = \log \sum_{z_{1:T}} p_\theta(x_{1:T}, z_{1:T}),$$

evaluated via the forward algorithm with scaling to prevent numerical underflow. Parameters are estimated via maximum likelihood using the EM (Baum–Welch) algorithm (7). Because the likelihood surface is non-convex and may contain singularities, estimation is repeated across multiple random initializations.

Hyperparameters and Model Variants

The primary hyperparameter is the number of hidden states K. In the empirical experiments I restrict attention to diagonal covariance emission models to maintain computational feasibility in the high-dimensional setting.

For each value of K, the EM algorithm is run R=5 times with independent initializations. The model achieving the highest log-likelihood among restarts is retained for comparison. Computational complexity per EM iteration for the diagonal covariance models used in this study scales approximately as

$$O(TK^2 + TKd),$$

reflecting the interaction between the number of hidden states K, the observation dimension d, and the sample size T.

3.2 Data and Feature Construction

The empirical analysis employed high-frequency limit order book data in the form of Nasdaq TotalView–ITCH in MBP-10 format. The observations were indexed in event time and ordered according to the event timestamp tsevent. Before the feature construction, rows containing missing values in the necessary price or size columns were dropped, and rows containing invalid book states where the bid price was greater than or equal to the ask price were discarded.

Two different feature constructions were analyzed in the paper. First, the 3D representation was derived from top-of-book information. In this case, the feature vector was composed of the mid-price change, the best-level order flow imbalance, and the inverse best-level depth for each event in the dataset. The mid-price was calculated as the average of the best bid and best ask prices, and the mid-price change was calculated by first-differencing the mid-price.

The second feature construction was 20D, derived from the first ten levels of the order book. In this case, the level-specific order flow imbalance feature and the level-specific inverse depth feature were calculated for each level in the order book, which resulted in 10 order flow imbalance features and 10 inverse depth features. Again, rows containing missing values in any of the selected features were discarded, and only events containing strictly positive depth in all selected levels were included in the dataset.

Finally, each dataset was standardized according to the z-score method. In order to keep the analysis computationally tractable while still preserving a large amount of observations in the dataset, the first 50,000 events were used in the analysis in each case. In other words, the final observation dimensions used in the analysis were $d = 3$ and $d = 20$.

3.3 Model Selection Criteria

For each fitted model, let $\hat{\theta}$ denote the parameter estimate obtained from the EM algorithm with the highest log-likelihood among multiple random initializations. The corresponding observed-data log-likelihood $\ell(\hat{\theta})$ is then used to compute the model selection criteria.

$$AIC = -2\ell(\hat{\theta}) + 2p,$$

$$BIC = -2\ell(\hat{\theta}) + p \log T,$$

where p is the number of free parameters in the model. For a Gaussian-emission HMM with diagonal covariance matrices,

$$p = (K - 1) + K(K - 1) + Kd + Kd,$$

corresponding to the initial state probabilities, transition matrix parameters, emission means, and diagonal covariance parameters.

The criteria are evaluated across a grid of candidate models with $K \in \{1, \ldots, 20\}$. The resulting curves are analyzed to detect monotonic decreases, interior minima, and possible divergence between AIC and BIC.

3.4 Stability and Redundancy Diagnostics

To evaluate whether selected models correspond to stable latent structure (Hypotheses 1–3), several diagnostic procedures are applied.

(A) Initialization Stability

For each K, let $\{\hat{\theta}^{(r)}\}_{r=1}^{R}$ denote EM solutions obtained from different random initializations. Because HMM likelihoods are invariant to permutations of state labels, the estimated states are aligned across runs using the Hungarian algorithm (6) based on emission means.

After alignment, the following quantities are computed:

• Frobenius norm differences between transition matrices

• Pairwise KL divergence between emission distributions (10)

• Fraction of time points where aligned MAP state sequences disagree

High variability across restarts indicates multimodality and weak identifiability of the fitted model.

(B) Parameter Redundancy

To detect redundant regimes, pairwise distances between emission distributions are computed using the Kullback–Leibler divergence:

$$D_{k\ell} = \mathrm{KL}\left(\mathcal{N}(\mu_k, \Sigma_k) \parallel \mathcal{N}(\mu_\ell, \Sigma_\ell)\right).$$

States with small KL divergence relative to typical inter-state distances and similar transition behavior are interpreted as potentially redundant regimes, providing evidence of model over-selection.

3.5 Implementation and Computational Resources

The full implementation used in this project is publicly available at

`https://github.com/ghalebkhalil/cse515-hmm-model-selection`.

All algorithms were implemented from scratch in Python using NumPy and SciPy. In particular, the forward recursion, backward recursion, smoothing step, Baum–Welch updates, model-selection pipeline, and stability diagnostics were coded directly from the mathematical formulations in (1; 2), without relying on external HMM packages. Experiments are conducted on standard CPU hardware. Computational feasibility is maintained by: Restricting K to a moderate range, Limiting subsample sizes, Using scaled forward recursion.

Consistent with the project scope, the methodology does not derive new information criteria. Instead, it evaluates whether established model selection tools remain reliable in high-dimensional, non-regular latent-state settings typical of market microstructure data (5).

4 Results / Summary

`https://github.com/ghalebkhalil/cse515-hmm-model-selection`.

This section presents the main empirical findings of the study. We show that likelihood-based model selection criteria, log-likelihood, AIC, and BIC, consistently favor increasingly complex models as the number of hidden states K increases, without identifying a meaningful optimal value. We then examine the internal structure of these high-K models to understand how the additional states are utilized.

Two experimental setups were considered in this work. In one setup, a three-dimensional observation vector is used, comprising components such as change in mid-price, order flow imbalance, and inverse best level depth. In another setup, a twenty-dimensional feature vector is considered, comprising order flow imbalance and inverse depth features at ten price levels in the order book. In both experimental setups, the models are fitted using the Baum-Welch Expectation-Maximization algorithm (7) with diagonal covariances and multiple random restarts.

For each experimental setup, the number of hidden states in the model is varied over a grid, and the maximized log likelihood is computed along with other model selection criteria such as AIC (3) and BIC (4). Subsequently, a high-K model is identified using these criteria. Finally, diagnostics on the identified model are performed to explore how the additional states in the high-K model are utilized by the model.

The results are grouped into two sets. The first set explores how well likelihood-based model selection criteria perform in identifying the number of hidden states in Gaussian Hidden Markov Models when K is large. The second set explores how well additional states are utilized in high-K models.

4.1 Result 1: Behavior of likelihood-based model selection

This result directly addresses Hypothesis 1, which tests whether likelihood-based criteria lead to over-selection. The first result concerns the behavior of maximized log likelihood and information criteria as a function of increasing numbers of hidden states. In both the $3D$ and $20D$ experiments, it is seen that the log likelihood on the observed data increases smoothly with increasing values of K over the entire range. This is expected, as more hidden states imply a more flexible model. What is more interesting is that this improvement continues even for large values of K. In both experiments, it is also seen that the information criteria, AIC and BIC, have a monotonic decreasing trend with increasing values of K. In both cases, the minimum values are obtained when K is set to its highest value. In the high-dimensional experiment, both AIC and BIC converge to $K = 20$, which is the upper bound on our grid.

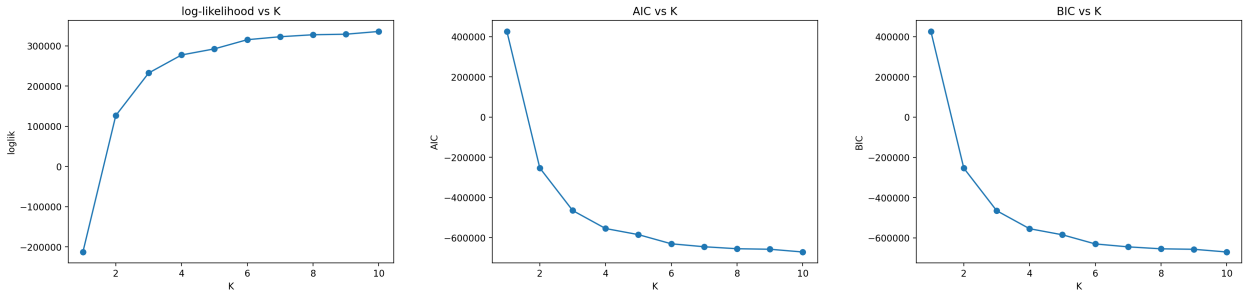
Table 1 summarizes the model-selection outcome for the 20-dimensional experiment.

Table 1: Model-selection outcome for the 20-dimensional experiment.

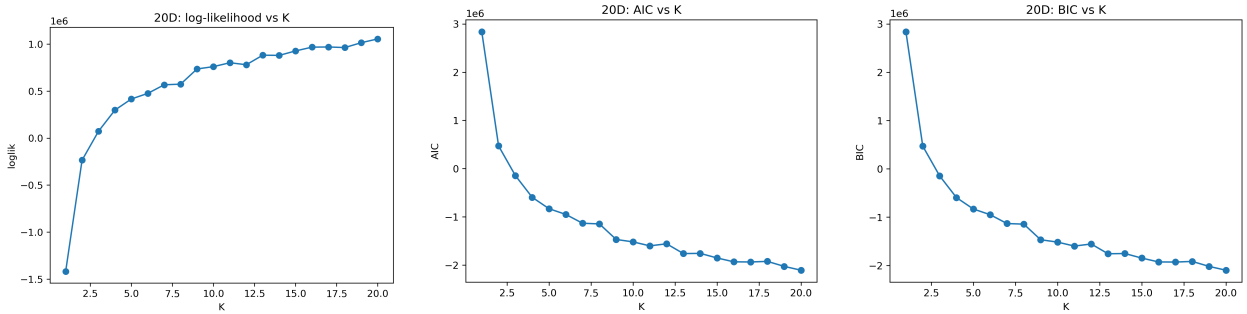
K selected by AIC	K selected by BIC	Log-likelihood	AIC	BIC
20	20	1,055,786.44	-2,109,174.87	-2,098,599.96

These results show that the information criteria continue to favor increasingly complex HMMs over the entire range of candidate models that were examined. The fact that both AIC and BIC select K = 20, the upper bound of the search grid, indicates that neither criterion identifies a finite optimal model size within the explored range.

Figure 1 shows the corresponding curves for log-likelihood, AIC, and BIC as a function of K. The log-likelihood grows steadily with the number of states, while both information criteria decrease without forming a clear interior minimum.



(a) 3D: log-likelihood vs. K (b) 3D: AIC vs. K (c) 3D: BIC vs. K



(d) 20D: log-likelihood vs. K (e) 20D: AIC vs. K (f) 20D: BIC vs. K

Figure 1: Model-selection curves for the 3-dimensional and 20-dimensional experiments. In both settings, the maximized log-likelihood increases as the number of hidden states K grows, while AIC and BIC decrease across the scanned range rather than forming a clear interior minimum.

As shown in Figure 1(a,d), the log-likelihood increases monotonically over the entire range $K \in \{1, \ldots, 20\}$, with no observable saturation. Figures 1(b,c,e,f) show that both AIC and BIC decrease monotonically as K increases, with no interior minimum observed in either the 3D or 20D setting. These experiments directly address the scope of the project,

which investigates whether classical information criteria provide reliable guidance for selecting the number of hidden regimes in Gaussian HMMs applied to high-frequency financial data.

4.2 Result 2: Structure of the selected high-K models

This result addresses Hypothesis 3, which examines whether increasing K leads to fragmentation of the latent state structure. After identifying the preferred model size according to the information criteria, I examined the internal structure of the selected models to understand how the additional states are used.

A first diagnostic is the state occupancy distribution, which measures how frequently each hidden state is visited in the smoothed posterior state probabilities. For the 3-dimensional experiment with $K = 19$, the most frequently visited state accounts for approximately 31% of the observations, while the least frequent state accounts for less than 1%. A similar pattern appears in the 20-dimensional experiment with $K = 20$, where the largest state occupies roughly 33% of the sample and the smallest state about 1%.

Figure 2 shows the occupancy distribution for the 20-state model. The distribution is highly uneven, with a small number of states capturing large portions of the data and several states representing much smaller subsets.

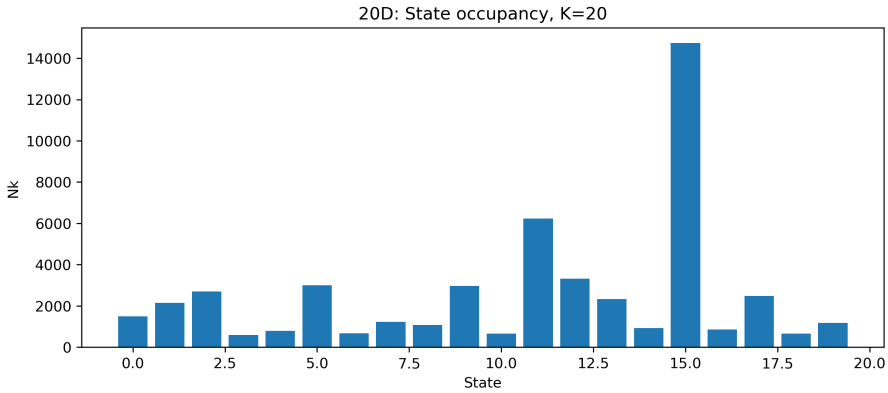


Figure 2: State occupancy for the selected 20-dimensional model with $K = 20$. The distribution is highly uneven: a small number of states account for a large share of the observations, while several states have much smaller effective mass.

I also examined the transition matrices and MAP regime sequences for the selected models. The transition matrices contain several states with high self-transition probabilities together with substantial movement across many other states. The corresponding regime paths show frequent switching between multiple states over short time horizons. These findings provide empirical evidence that the likelihood surface is highly multimodal, and that information-criterion-based model selection may favor solutions that are not stable across repeated estimations. As shown in Figure 3, several states exhibit high self-transition probabilities (visible along the diagonal), while non-negligible transition mass is distributed across many off-diagonal entries.

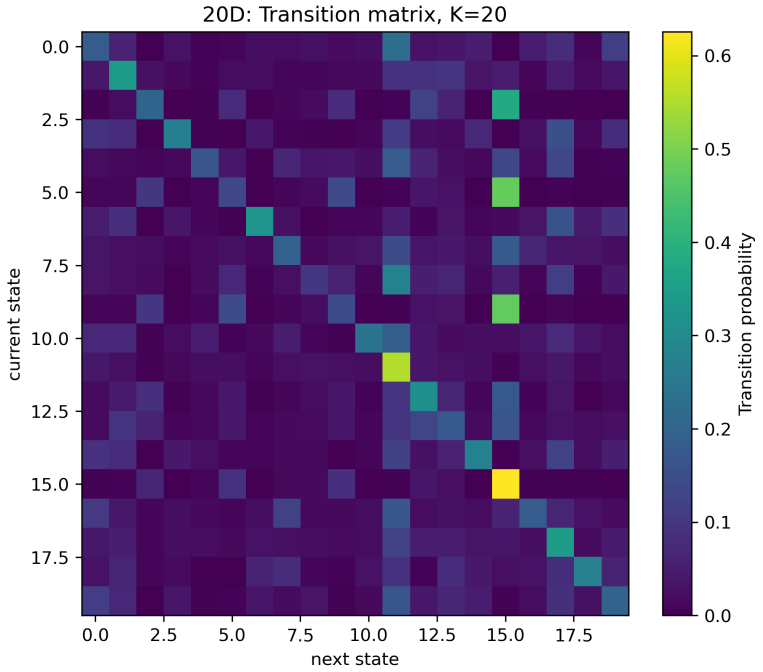


Figure 3: Estimated transition matrix for the selected 20-dimensional model. Some states show strong persistence, while substantial transition mass remains spread across multiple other states.

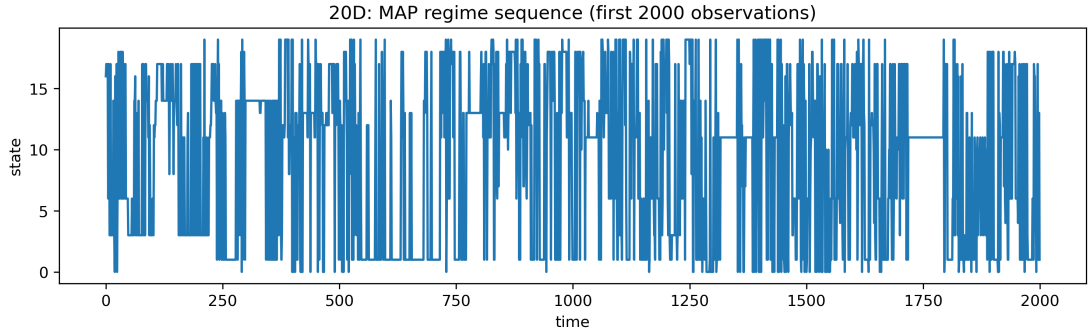


Figure 4: MAP regime sequence for the first 2,000 observations in the selected 20-dimensional model. The fitted path switches among many latent states over short horizons.

Finally, I evaluated the pairwise separation of emission distributions using symmetric KL divergence between the Gaussian emission components. In the 20-dimensional experiment no state pairs satisfied the redundancy threshold used in the diagnostic procedure. This indicates that the fitted Gaussian components are not near duplicates under the chosen distance measure.

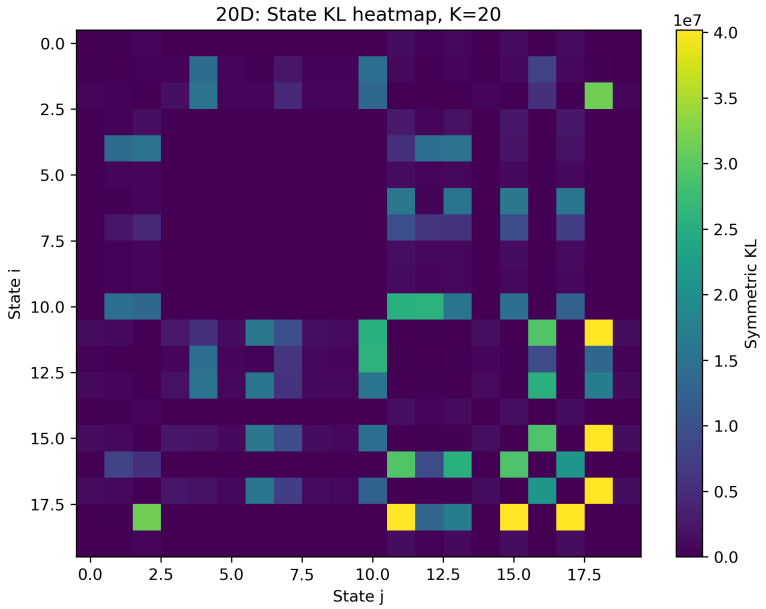


Figure 5: Symmetric KL divergence heatmap between the Gaussian emission distributions in the selected 20-dimensional model. The distances are heterogeneous across state pairs, indicating that separation between fitted states is uneven.

Together, these diagnostics describe how the large-K models partition the observation space and how the inferred hidden states are used across the dataset.

5 Discussion

This study evaluates whether classical likelihood-based model selection criteria, namely AIC and BIC, provide reliable guidance for selecting the number of hidden regimes in Gaussian Hidden Markov Models applied to high-frequency financial data. The empirical results show that as the number of hidden states K increases, the maximized log-likelihood increases monotonically, while both AIC and BIC decrease without exhibiting a clear interior minimum. In both experimental settings, the criteria select the largest model considered, indicating a systematic preference for increasingly complex models.

To better understand this behavior, we examined structural diagnostics of the selected high-K models, including state occupancy distributions, transition matrices, MAP regime sequences, and pairwise KL divergence between emission distributions. These diagnostics reveal that a small number of states capture a large proportion of the observations, while many states have very low effective probability. This indicates that increasing K leads to a fragmented representation of the observation space, where additional states capture fine statistical variations rather than distinct underlying regimes.

A key implication is that improvements in likelihood do not necessarily correspond to improvements in the interpretability or stability of the inferred regime structure.

These findings suggest that classical likelihood-based criteria may not provide reliable guidance for selecting the number of hidden regimes in high-dimensional financial HMMs. Although these criteria are designed to balance model fit and complexity, in practice the continuous improvement in log-likelihood dominates the penalty terms, leading to a preference for increasingly complex models.

The results provide empirical support for the hypotheses formulated in the project scope. First, the experiments confirm likelihood over-selection, with AIC and BIC favoring larger models without identifying a finite optimum. Although only the best-performing run is retained for each value of K, the need for multiple random initializations, together with the observed variability in fitted structures across runs, provides empirical support for Hypothesis 2. In particular, these observations indicate that the likelihood surface is highly multimodal and that the resulting models are sensitive to initialization, leading to instability in the inferred regime structure. Third, diagnostic analysis of high-K models indicates fragmentation of hidden regimes, with uneven state occupancy and frequent switching between states. Together, these findings suggest that information criteria alone may not be sufficient for reliable model selection in high-dimensional financial HMMs.

5.1 What was Easy

Some aspects of this project were more tractable than others, with the interpretation of the results, as well as the creation of the visualizations, proving relatively easier after the successful fitting of the models. For example, the creation of the likelihood plots with respect to the varying values of K, as well as the state occupancy plots and heatmaps, could be achieved relatively quickly using the tools available in the libraries used in the project. The plots provided good qualitative insights into the behavior of the selection criteria used in the models, as well as the state occupancy distributions.

Interpreting the trends in the results obtained from the models was also relatively intuitive, with the monotonically increasing nature of the log-likelihood, the monotonically decreasing nature of the AIC and BIC values, as well as the imbalance in the state occupancy distributions, not requiring further modeling complexities beyond the interpretation of the values obtained from the models. Once the results were obtained from the models, drawing broad inferences on the issues of over-selection as well as regime fragmentation was relatively straightforward from the trends obtained in the results.

Thus, while the underlying modeling framework was relatively complex, the interpretation of the models, as well as the creation of the visualizations, was relatively easier to implement, providing good insights into the behavior of the models used in the project.

5.2 What was Difficult

The major problem with this project has been the implementation of the entire Hidden Markov Model (HMM) pipeline from the ground up. Although the theoretical foundation is well established, the translation of the theory into a reliable and accurate implementation required close attention to a variety of sources. In particular, the scaled forward-backward algorithm, the accurate computation of the log-likelihood, and the derivation of accurate expectation-maximization (EM), also known as the Baum-Welch algorithm, updates across all stages of the algorithm required significant work to ensure numerical stability and accuracy.

Another major problem with the project has been the computational expense associated with the implementation of the algorithm across a range of different hidden states, K. The non-convexity of the log-likelihood function and the increase in the dimensionality of the parameter space with increasing K meant that the procedure had to be repeated a number of times. This made the overall procedure computationally intensive.

A final problem with the project has been the interpretation of the results of the algorithm with a large number of states. Although the results of the algorithm with a large number of states were able to improve the log-likelihood, the results were often fragmented and unbalanced, with a large number of states and complex transition structures. In particular, the question of whether the additional states provided any additional information required additional diagnostics, including the computation of the KL divergence between the emission distributions. However, the emission distributions themselves also raise a number of interpretational difficulties.

5.3 Recommendations for Future Work

There are various avenues for extension of this research. First, different model selection methodologies might be considered to overcome some of the problems with AIC and BIC. For example, cross-validation or predictive likelihood might offer better guidance on choosing the optimal number of hidden states for complex models. Second, additional regularization methodologies might be used to counterbalance the model's tendency to allocate many states to small data sets. Penalized likelihood might offer better guidance on model parsimony and potentially improve interpretability. Third, future research might investigate whether this phenomenon occurs with different data sets or different representations of limit order book data. This research utilizes a specific data representation, and it is possible that different results might be found with different data representations due to their different dimensionality and statistical characteristics.

Overall, this research demonstrates the importance of combining information criterion with diagnostics in choosing HMMs. It is possible that information criteria alone might not provide sufficient information on the model's structure, and additional diagnostics might provide useful insights into how the model maps onto the data.

References

[1] L. R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.

[2] O. Cappé, E. Moulines, and T. Rydén. *Inference in Hidden Markov Models*. Springer, 2005.

[3] H. Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974.

[4] G. Schwarz. Estimating the dimension of a model. *Annals of Statistics*, 6(2):461–464, 1978.

[5] R. Cont, A. Kukanov, and S. Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 9(1):47–88, 2011.

[6] H. W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955.

[7] A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society Series B*, 39(1):1–38, 1977.

[8] K. P. Murphy. *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012.

[9] C. M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[10] S. Kullback and R. A. Leibler. On information and sufficiency. *Annals of Mathematical Statistics*, 22(1):79–86, 1951.

[11] Z. Ghahramani. An introduction to hidden Markov models and Bayesian networks. *International Journal of Pattern Recognition and Artificial Intelligence*, 15(1):9–42, 2001.

[12] E. Seneta. *Non-negative matrices and Markov chains*. Springer, 2006.